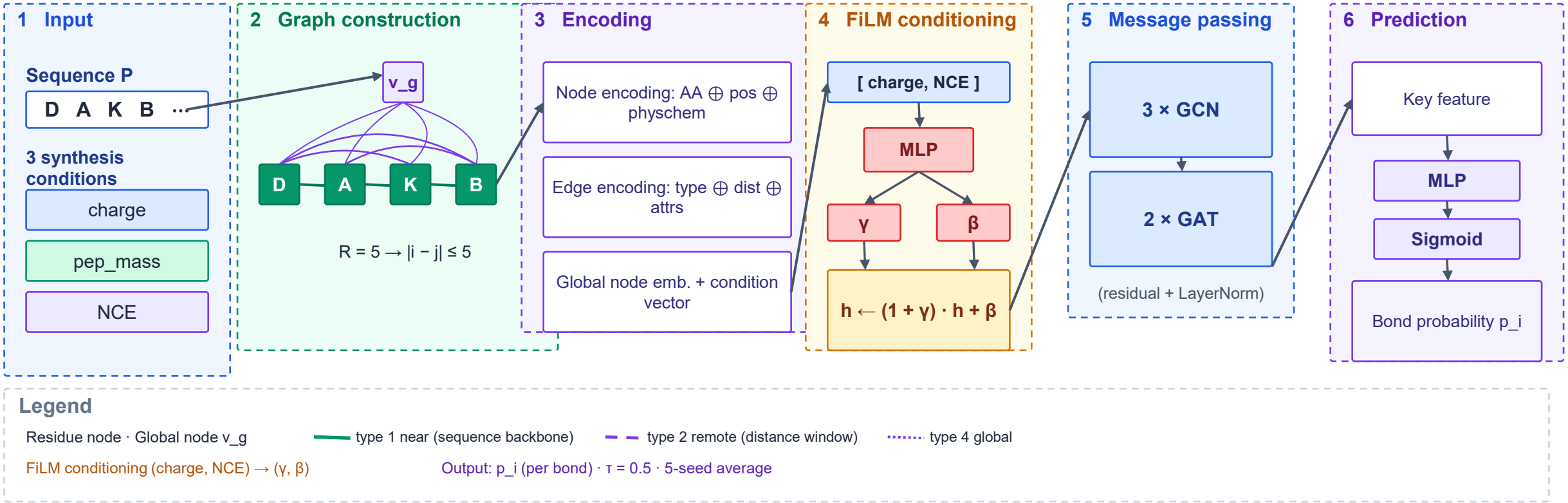

1 Input
Sequence P
D A K B ···
3 synthesis conditions
charge
pep_mass
NCE
2 Graph construction
v_g
D
A
K
B
R = 5 → |i − j| ≤ 5
3 Encoding
Node encoding: AA ⊕ pos ⊕ physchem
Edge encoding: type ⊕ dist ⊕ attrs
Global node emb. + condition vector
4 FiLM conditioning
[charge, NCE]
MLP
γ
β
h ← (1 + γ) · h + β
5 Message passing
3 × GCN
2 × GAT
(residual + LayerNorm)
6 Prediction
Key feature
MLP
Sigmoid
Bond probability p_i
Legend
Residue node · Global node v_g
type 1 near (sequence backbone)
type 2 remote (distance window)
type 4 global
FiLM conditioning (charge, NCE) → (γ, β)
Output: p_i (per bond) · τ = 0.5 · 5-seed average